

# MALAYSIA BID ROUND 2026 (MBR 2026)

## BID PROPOSAL — EXPLORATION BLOCKS & DISCOVERED RESOURCE OPPORTUNITIES

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### 1. EXECUTIVE SUMMARY

PETRONAS, through Malaysia Petroleum Management (MPM), launched the Malaysia Bid Round 2026 (MBR 2026) on 10 February 2026 under the theme “**Advantaged Energy: Accelerating and Shaping Tomorrow.**” The round offers **15 opportunities** — 9 exploration blocks and 6 Discovered Resource Opportunities (DROs) — across three major basin provinces: the mature **Malay Basin**, the emerging **West Sarawak Basin**, and the frontier **Sandakan Basin**.

#### Key Parameters

| Parameter           | Value                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------|
| Total Opportunities | 15 (9 exploration + 6 DRO)                                                                   |
| Basins              | Malay, West Sarawak, Sandakan                                                                |
| Fiscal Terms        | Enhanced Profitability Terms (EPT) PSC for exploration; Small Field Asset (SFA) PSC for DROs |
| Bid Deadline        | 1 October 2026 (all except SB304: 30 June 2027)                                              |
| PSC Award Target    | February 2027                                                                                |
| PSC Signing Target  | April 2027                                                                                   |
| Prerequisite        | PETRONAS myPROdata subscription                                                              |
| Data Rooms          | Open since 10 February 2026                                                                  |

#### Recommendation Summary

**RECOMMENDED TO BID:** - **PM447** — Malay Basin, near-field with new play ideas (Pre-Tertiary carbonates) - **PM448** — Malay Basin, near-field, shallow Pre-Triassic targets - **Cempaka Cluster (DRO)** — Small field, ready-to-develop, low-risk

**CONSIDER WITH PARTNERS:** - **SK404** — West Sarawak, high potential but requires significant capital - **SK331** — West Sarawak, emerging play with 3D seismic data - **Enggang Cluster (DRO)** — Proven fields, requires redevelopment expertise

**NOT RECOMMENDED:** - **SB304** — Sandakan Basin frontier, extreme geological risk, no 3D seismic - **PM519/PM520** — Deep Pre-Tertiary targets, very high depth to basement risk

## 2. MBR 2026 PORTFOLIO OVERVIEW

### 2.1 Exploration Blocks

| Block        | Basin      | Area (km <sup>2</sup> ) | Water Depth (m) | PSC Terms | Minimum Work Commitment                                                   |
|--------------|------------|-------------------------|-----------------|-----------|---------------------------------------------------------------------------|
| <b>PM440</b> | Malay      | 4,840                   | 50–60           | EPT       | Phase 1: 3 yrs, drill 1 well                                              |
| <b>PM447</b> | Malay      | 4,325                   | 20–60           | EPT       | Phase 1: 3 yrs, drill 1 well + license MC3D                               |
| <b>PM448</b> | Malay      | —                       | —               | EPT       | Repro<br>Phase 1: 3 yrs, drill 1 well                                     |
| <b>PM519</b> | Malay      | 5,629                   | 5–75            | EPT       | Phase 1: 4 yrs, drill 1 well + 2D                                         |
| <b>PM520</b> | Malay      | 6,110                   | 5–75            | EPT       | OB<br>Phase 1: 4 yrs, drill 1 well + 3D                                   |
| <b>SK330</b> | W. Sarawak | 5,038                   | 120–300         | EPT       | TZ seismic<br>Phase 1: 3 yrs, drill 1 well + license Beicip               |
| <b>SK331</b> | W. Sarawak | 2,835                   | 70–200          | EPT       | MC PH2<br>Phase 1: 3 yrs, drill 1 well + license Beicip                   |
| <b>SK404</b> | W. Sarawak | 4,829                   | 40–100          | EPT       | MC PH2<br>Phase 1: 3 yrs, drill 1 well + license Beicip                   |
| <b>SB304</b> | Sandakan   | 7,030                   | 0–20            | EPT       | MC PH1<br>Phase 1: 3 yrs, acquire 500 km <sup>2</sup> 3D OBN + 1000 km 2D |

### 2.2 Discovered Resource Opportunities (DRO)

| DRO                    | Location      | Fields                                                                   | Fiscal Term |
|------------------------|---------------|--------------------------------------------------------------------------|-------------|
| <b>Cempaka Cluster</b> | Pen. Malaysia | Saujana, Serok, Semangkok NAG, Semangkok Timur                           | SFA PSC     |
| <b>Teratai Cluster</b> | Pen. Malaysia | Sotong, Malong, Anding, Aji, Aji Sentang, Parang, Naga Kecil, Naga Dalam | SFA PSC     |
| <b>Enggang Cluster</b> | Sarawak       | Salam, Benum, Patawali, Gagau                                            | SFA PSC     |

| DRO                  | Location | Fields                                           | Fiscal Term |
|----------------------|----------|--------------------------------------------------|-------------|
| <b>NURI Funnel</b>   | Sarawak  | Nuang, Rompin, KT1, Endau, Mahkota E2, Serunai   | SFA PSC     |
| <b>Nilam Cluster</b> | Sabah    | Trusmadi, Madalon, Papar, Padas, Grafit          | SFA PSC     |
| <b>Zamrud Funnel</b> | Sabah    | Tembungo, SW Emerald, Danum, Rusa Timur, Sipadan | SFA PSC     |

### 3. FISCAL TERMS ANALYSIS

#### 3.1 Enhanced Profitability Terms (EPT) — Exploration Blocks

The EPT PSC represents PETRONAS's modern fiscal framework for shallow water exploration, introduced in 2021 to replace the legacy Revenue over Cost (R/C) structure.

| Component                    | Terms                                                                    |
|------------------------------|--------------------------------------------------------------------------|
| <b>Cash Payment</b>          | 10% of gross production to Federal/State Governments                     |
| <b>Cost Recovery Ceiling</b> | Fixed 70% of gross production                                            |
| <b>Profit Sharing</b>        | Based on Profitability Index (PI = cumulative revenue ÷ cumulative cost) |
| <b>PI 1.5</b>                | Contractor receives 90% of profit oil                                    |
| <b>1.5 &lt; PI 2.5</b>       | Contractor share interpolated 90% → 30%                                  |
| <b>PI &gt; 2.5</b>           | Contractor receives 30% of profit oil                                    |

**Assessment:** The EPT terms are highly competitive globally. The fixed 70% cost recovery ceiling provides certainty, and the PI-based profit sharing rewards early exploration success. At low PI (early investment recovery), the contractor retains 90% — this is excellent for frontier exploration.

#### 3.2 Small Field Asset (SFA) PSC — DROs

| Component                     | Terms                                                                        |
|-------------------------------|------------------------------------------------------------------------------|
| <b>Cash Payment</b>           | 10% of gross production                                                      |
| <b>PETRONAS Share ( % )</b>   | Minimum set, biddable item, fixed throughout contract                        |
| <b>Contractor Share ( % )</b> | % of gross production includes ALL costs (Capex, Opex, Abex) + profit margin |
| <b>Equation</b>               | % + % = 90% (10% government cash payment)                                    |

**Assessment:** The SFA terms are deliberately simplified to attract smaller players. The biddable % means competitive bidding will drive PETRONAS's share. The all-inclusive % structure reduces fiscal complexity — ideal for small field development where cost control is paramount.

#### 3.3 Comparative Analysis

| Fiscal Feature | Legacy R/C PSC      | EPT PSC   | SFA PSC    |
|----------------|---------------------|-----------|------------|
| Cash Payment   | 10%                 | 10%       | 10%        |
| Cost Recovery  | Variable (30–70%)   | Fixed 70% | Simplified |
| Profit Sharing | R/C ratio dependent | PI-based  | Biddable / |
| Complexity     | High                | Medium    | Low        |

| Fiscal Feature | Legacy R/C PSC | EPT PSC     | SFA PSC      |
|----------------|----------------|-------------|--------------|
| Best For       | Large fields   | Exploration | Small fields |

## 4. EXPLORATION BLOCKS — GEOLOGICAL ASSESSMENT

### 4.1 MALAY BASIN BLOCKS (PM440, PM447, PM448, PM519, PM520)

**4.1.1 Regional Geology** The Malay Basin is a **Cenozoic rift basin** in the southern South China Sea, ~500 km long × 200 km wide, trending NW-SE. It is Malaysia’s most prolific hydrocarbon province, contributing **~40% of national resources** with cumulative discoveries exceeding **12 billion barrels of oil equivalent** (USGS, 1999).

**Tectonic Evolution:** 1. **Late Eocene–Oligocene (Syn-rift):** Extensional half-grabens filled with lacustrine shales and continental clastics (Groups M, L, K) 2. **Early Miocene (Sag/Transition):** Fluvial-deltaic systems prograde into the basin (Groups J, I, H) 3. **Middle–Late Miocene (Inversion):** Transpressional inversion creates E-W trending anticlines — the primary structural traps (Groups G, F, E, D) 4. **Pliocene–Recent (Post-inversion):** Marine transgression, regional subsidence (Groups A, B, C)

**Petroleum Systems:** - **Source Rocks:** Oligocene–Miocene lacustrine shales (Groups L, M) and Miocene coaly shales (Groups H, I). Two source types: Type I/II lacustrine kerogen and Type III terrestrial/coaly kerogen. - **Reservoirs:** Fluvial-deltaic sandstones (Groups D, E, F, H, I) with porosities 15–25%. Deep Pre-Tertiary carbonates represent an emerging play. - **Seals:** Basin-wide marine shales (K shale is regionally extensive), intra-formational shales. - **Traps:** Inversion anticlines (E-W trending), fault-dependent closures, stratigraphic traps. - **Migration:** Vertical along deep-seated faults, lateral along carrier beds.

**Key Insight:** The basin is **mature for conventional plays**. Remaining Yet-to-Find (YTF) estimated at **~2–3 Bboe** (Madon, 2021), primarily in: - Deep Pre-Tertiary carbonates (new play concept) - Stratigraphic traps in Groups F–I (requires advanced seismic) - Basin-flank plays on the NE ramp margin

#### 4.1.2 Block PM440 — Near-Field Static

| Parameter         | Value                             |
|-------------------|-----------------------------------|
| Area              | 4,840 km <sup>2</sup>             |
| Water Depth       | 50–60 m                           |
| PSC               | EPT                               |
| MWC               | Phase 1: 3 years, drill 1 well    |
| Data              | myPro 3D MC3D Repro (3,000 sq km) |
| MPM Prospectivity | Near-field, static play           |

**Geological Narrative:** PM440 is positioned in the **central Malay Basin** — a mature area with proven petroleum systems. The block sits within the main oil-prone fairway where Groups H and I source rocks are in the oil window. The near-field setting means existing infrastructure (platforms, pipelines) may be accessible for early monetization.

**Play Types:** - Conventional Miocene clastic inversion anticlines (proven) - Potential stratigraphic traps in deltaic systems - Possible Pre-Tertiary carbonate targets at depth

**Risk Assessment:** - **Source:** LOW — proven regional kitchen - **Reservoir:** MEDIUM — conventional targets well-understood, but quality varies - **Trap:** MEDIUM — need to identify undrilled closures - **Seal:** LOW — K shale regionally effective - **Charge:** LOW — active kitchen

**BID RECOMMENDATION: CONDITIONAL BID** — if 3D seismic data room confirms undrilled structural closures with >50 MMboe potential.

#### 4.1.3 Block PM447 — New Play Ideas

| Parameter         | Value                                                       |
|-------------------|-------------------------------------------------------------|
| Area              | 4,325 km <sup>2</sup>                                       |
| Water Depth       | 20–60 m                                                     |
| PSC               | EPT                                                         |
| MWC               | Phase 1: 3 years, drill 1 well + license MC3D Repro         |
| Data              | Mypro 3D MC3D Repro (3,000 sq km)                           |
| MPM Prospectivity | Portfolio being interpreted on recently acquired Penyu MC3D |

**Geological Narrative:** PM447 is a **high-interest block** because MPM has identified it as hosting **new play ideas** beyond the conventional Miocene clastic fairway. The block is positioned where the **Penyu Basin** meets the Malay Basin, creating a unique geological junction. Recent MC3D seismic acquisition (Penyu MC3D) has been completed and is being interpreted — this suggests MPM sees material upside.

**Play Types:** - **Shallow Pre-Triassic carbonate play** — MPM specifically references this as a target - Conventional Miocene clastic inversion anticlines - Possible syn-rift clastic plays in deeper Groups K, L, M

**Risk Assessment:** - **Source:** LOW-MEDIUM — Penyu Basin source rocks less proven than central Malay Basin - **Reservoir:** MEDIUM — Pre-Tertiary carbonates are fracture-dependent (3–8% porosity base) - **Trap:** MEDIUM — carbonate buildups may be structurally complex - **Seal:** MEDIUM — need regional seal above carbonate section - **Charge:** MEDIUM — migration pathways from Malay Basin kitchen uncertain

**BID RECOMMENDATION: BID** — The Pre-Tertiary carbonate play represents a **material new play concept** in a mature basin. The MC3D Repro data reduces technical risk. If carbonates deliver, upside could be >100 MMboe per structure.

#### 4.1.4 Block PM448 — Shallow Pre-Triassic

| Parameter         | Value                                         |
|-------------------|-----------------------------------------------|
| Area              | —                                             |
| Water Depth       | —                                             |
| PSC               | EPT                                           |
| MWC               | Phase 1: 3 years, drill 1 well                |
| Data              | myPro 3D available                            |
| MPM Prospectivity | Shallow Pre-Triassic opportunity, drill-ready |

**Geological Narrative:** PM448 is described as having a “**drill-ready Shallow Pre-Triassic opportunity**” — this is significant because it means MPM has already identified a specific prospect and the block is positioned for near-term drilling. The Pre-Tertiary carbonate play in the Malay Basin is analogous to proven plays in the Penyu Basin and represents a step-out from conventional Miocene targets.

**Play Types:** - **Pre-Tertiary carbonate** — primary target, shallow enough for efficient drilling - Conventional Miocene clastic (secondary)

**Risk Assessment:** - **Source:** MEDIUM — Pre-Tertiary charge model needs validation - **Reservoir:** MEDIUM — carbonate porosity depends on fracture development and dissolution - **Trap:** LOW-MEDIUM — drill-ready suggests structural closure mapped on seismic - **Seal:** MEDIUM — Pre-Tertiary seal integrity needs verification - **Charge:** MEDIUM — migration from deeper kitchen possible

**BID RECOMMENDATION: BID** — Drill-ready status and low MWC (1 well) make this a high-value target. Even a moderate discovery (30–50 MMboe) would be commercial under EPT terms.

#### 4.1.5 Block PM519 — Deep Pre-Tertiary

| Parameter         | Value                                                                |
|-------------------|----------------------------------------------------------------------|
| Area              | 5,629 km <sup>2</sup>                                                |
| Water Depth       | 5–75 m                                                               |
| PSC               | EPT                                                                  |
| MWC               | Phase 1: 4 years, drill 1 well + 2D OBN                              |
| Data              | myPro 3D negligible; Penyu MC3D 1,300 sq km                          |
| MPM Prospectivity | Pre-Tertiary opportunities relatively shallow in PM519 (1–2 km dbml) |

**Geological Narrative:** PM519 hosts Pre-Tertiary targets that MPM notes are “**relatively shallow**” at 1–2 km below mud line — this is drillable with conventional jack-up rigs. However, the data availability is described as “negligible” for myPro 3D, with only Penyu MC3D (1,300 km<sup>2</sup>) covering a portion of the block.

**Play Types:** - Pre-Tertiary carbonate/clastic — primary target - Conventional Miocene clastic — secondary

**Risk Assessment:** - **Source:** MEDIUM-HIGH — Pre-Tertiary charge model speculative - **Reservoir:** HIGH — fracture porosity (3–8%) may not be sufficient - **Trap:** MEDIUM — structural mapping possible on available seismic - **Seal:** HIGH — Pre-Tertiary seal integrity uncertain - **Charge:** HIGH — migration pathways unclear

**BID RECOMMENDATION: NO BID** — Poor data coverage (“negligible” 3D), combined with high geological risk on Pre-Tertiary targets. The 4-year Phase 1 with 2D OBN acquisition adds cost. Better opportunities exist in PM447 and PM448.

#### 4.1.6 Block PM520 — Deep Pre-Tertiary (>5 km)

| Parameter         | Value                                                    |
|-------------------|----------------------------------------------------------|
| Area              | 6,110 km <sup>2</sup>                                    |
| Water Depth       | 5–75 m                                                   |
| PSC               | EPT                                                      |
| MWC               | Phase 1: 4 years, drill 1 well + 500 sq km 3D TZ seismic |
| Data              | Negligible                                               |
| MPM Prospectivity | Pre-Tertiary targets very deep (>5 km dbml)              |

**Geological Narrative:** PM520 targets Pre-Tertiary plays at **>5 km below mud line** — this puts the targets in the **high-pressure, high-temperature (HPHT) regime** where drilling costs escalate dramatically and reservoir quality is severely compromised. The block also requires acquisition of 500 sq km of Transition Zone (TZ) 3D seismic — a significant upfront investment.

**Play Types:** - Pre-Tertiary carbonate/clastic at >5 km depth — extreme HPHT

**Risk Assessment:** - **Source:** HIGH — over-mature at these depths - **Reservoir:** VERY HIGH — extreme compaction and diagenesis at >5 km - **Trap:** HIGH — structural uncertainty at depth - **Seal:** HIGH — cap rock integrity under HPHT conditions - **Charge:** HIGH — timing and migration highly uncertain

**BID RECOMMENDATION: STRONG NO BID** — >5 km targets with negligible data and HPHT conditions represent **prohibitive technical and financial risk**. A single well could cost >\$50M with very low probability of success. This block requires technology that may not exist commercially.

## 4.2 WEST SARAWAK BASIN BLOCKS (SK330, SK331, SK404)

**4.2.1 Regional Geology** The West Sarawak Basin is an **emerging petroleum province** located offshore Sarawak, Malaysian Borneo. It forms part of the greater Sarawak Basin system that has produced hydrocarbons from the Baram Delta Province and Luconia Platform.

**Tectonic Setting:** - Complex **transtensional deformation** during the Tertiary related to India-Indochina collision - Sinistral strike-slip zones (Sabah Shear, West Baram-Tinjar Lines, Lupar Line) - NE-SW trending anticlines formed by compression and wrenching

**Petroleum Systems:** - **Source Rocks:** Coaly shales and organic-rich claystones of the Nyalau/Setap Formation (Oligocene–Miocene), TOC values 1.54–70 wt% - **Reservoirs:** Miocene deltaic and shallow marine sandstones (Miri/Belait/Seria Formations), Oligocene carbonates - **Traps:** NE-SW trending anticlines, fault-dependent closures, carbonate buildups - **Proven Plays:** Miocene clastic, Oligocene carbonate, Pre-Tertiary basement

**Key Insight:** West Sarawak is **less explored than the Malay Basin** but has proven hydrocarbon systems nearby. The challenge is **CO risk** in the Luconia area and the need for better understanding of source rock distribution and migration pathways.

#### 4.2.2 Block SK331 — West Sarawak Emerging

| Parameter         | Value                                                  |
|-------------------|--------------------------------------------------------|
| Area              | 2,835 km <sup>2</sup>                                  |
| Water Depth       | 70–200 m                                               |
| PSC               | EPT (2 phases)                                         |
| MWC               | Phase 1: 3 years, drill 1 well + license Beicip MC PH2 |
| Data              | myPro 3D (tbc sq km)                                   |
| MPM Prospectivity | Emerging West Sarawak                                  |

**Geological Narrative:** SK331 is positioned in the **emerging West Sarawak fairway** where recent exploration has shown encouraging results. The block benefits from Beicip multiclient study data (PH2), which provides regional geological context. Water depths of 70–200 m are manageable with modern jack-up or semi-submersible rigs.

**Play Types:** - Miocene clastic — deltaic and shallow marine sandstones - Oligocene carbonate — potential buildups - Possible stratigraphic traps

**Risk Assessment:** - **Source:** MEDIUM — coaly source rocks regionally present but distribution uncertain - **Reservoir:** MEDIUM — Miocene sandstones proven in nearby fields - **Trap:** MEDIUM — anticlinal structures mapped on seismic - **Seal:** MEDIUM — Miocene shales provide regional seal - **Charge:** MEDIUM-HIGH — migration pathways from kitchen areas need validation

**BID RECOMMENDATION: BID WITH PARTNER** — reasonable risk-reward if a partner with Sarawak exploration experience is secured. The Beicip study data reduces geological uncertainty.

#### 4.2.3 Block SK330 — Deepwater West Sarawak

| Parameter   | Value                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------|
| Area        | 5,038 km <sup>2</sup>                                                                                        |
| Water Depth | 120–300 m                                                                                                    |
| PSC         | EPT (2 phases)                                                                                               |
| MWC         | Phase 1: 3 years, drill 1 well + license Beicip MC PH2 + license 3,875 sq km TGS MC3D (\$5M USD license fee) |
| Data        | TGS MC3D 3,875 sq km                                                                                         |

**Geological Narrative:** SK330 has the **best seismic data coverage** of all MBR2026 blocks — 3,875 km<sup>2</sup> of TGS multiclient 3D seismic. However, the **\$5M USD data license fee** is a significant upfront cost. Water depths of 120–300 m require semi-submersible rigs or drillships, escalating well costs to \$40–60M per well.

**Play Types:** - Miocene clastic — deepwater fan and turbidite systems - Oligocene carbonate — potential reef buildups - Stratigraphic traps in deepwater setting

**Risk Assessment:** - **Source:** MEDIUM — same regional source kitchen - **Reservoir:** MEDIUM-HIGH — deepwater reservoir quality variable - **Trap:** MEDIUM — structural and stratigraphic traps possible - **Seal:** LOW-MEDIUM — deepwater shales effective - **Charge:** HIGH — long migration distances possible

**BID RECOMMENDATION: CONSIDER WITH CAUTION** — excellent data but high entry cost (\$5M license + deepwater well). Only viable for a well-capitalized operator with deepwater capability. Recommend farm-in strategy rather than direct bid.

#### 4.2.4 Block SK404 — West Sarawak Primary

| Parameter   | Value                                                  |
|-------------|--------------------------------------------------------|
| Area        | 4,829 km <sup>2</sup>                                  |
| Water Depth | 40–100 m                                               |
| PSC         | EPT (2 phases)                                         |
| MWC         | Phase 1: 3 years, drill 1 well + license Beicip MC PH1 |
| Data        | Mypro 3D (tbc) + Beicip MC PH1                         |

**Geological Narrative:** SK404 is the **most accessible West Sarawak block** — shallow water (40–100 m) allows jack-up rig operations at significantly lower cost than SK330. The Beicip multiclient PH1 study provides regional geological context. The shallow water setting also means proximity to existing Sarawak infrastructure.

**Play Types:** - Miocene clastic — deltaic and shallow marine - Oligocene carbonate — platform buildups - Potential near-field tie-back opportunities to existing Sarawak production

**Risk Assessment:** - **Source:** MEDIUM — regional source proven but local charge model needs work - **Reservoir:** MEDIUM — Miocene sandstones proven regionally - **Trap:** MEDIUM — structural closures expected - **Seal:** LOW-MEDIUM — regional shales effective - **Charge:** MEDIUM — shorter migration paths than SK330

**BID RECOMMENDATION: BID** — Best risk-reward ratio among West Sarawak blocks. Shallow water = lower drilling costs. Beicip study data available. Proximity to existing infrastructure. This is the **preferred entry point for West Sarawak exploration**.

### 4.3 SANDAKAN BASIN (SB304)

**4.3.1 Regional Geology** The Sandakan Basin is a **frontier basin** located offshore northeast Sabah. It is part of the greater Tarakan Basin system with proven hydrocarbons in nearby Indonesian waters (Badik, Aster, Tulip Fields — ~500 MMboe cumulative).

**Tectonic Setting:** - Located at the junction of the Sulu Sea and Celebes Sea - Complex tectonic history involving subduction, accretion, and strike-slip deformation - Miocene to Pliocene sedimentary fill overlying older basement

**Petroleum Systems:** - **Source Rocks:** Miocene and intra-Pliocene clastic source rocks (coaly and marine) - **Reservoirs:** Miocene to Eocene clastic, Oligocene carbonate, Pre-Tertiary basement - **Traps:** Structural (anticlines, fault blocks) and stratigraphic - **Proven Plays:** Miocene clastic, Oligocene carbonate, Pre-Tertiary

**Key Insight:** The Sandakan Basin is **truly frontier** — limited well control, no 3D seismic in SB304, and geological uncertainty is high. The positive: proven petroleum system in the greater Tarakan Basin, and MPM has extended the bid deadline to June 2027, suggesting they expect this block to attract patient capital.

#### 4.3.2 Block SB304 — Frontier Sandakan

| Parameter   | Value                         |
|-------------|-------------------------------|
| Area        | 7,030 km <sup>2</sup>         |
| Water Depth | 0–20 m (onshore to nearshore) |



| Parameter    | Value                                                                    |
|--------------|--------------------------------------------------------------------------|
| PSC          | EPT (2 phases)                                                           |
| MWC          | Phase 1: 3 years, acquire 500 sq km 3D OBN seismic + 1000 line-km 2D OBN |
| Data         | No 3D seismic available                                                  |
| Bid Deadline | 30 June 2027 (extended)                                                  |

**Geological Narrative:** SB304 is the **highest-risk, highest-reward block** in MBR2026. It covers 7,030 km<sup>2</sup> in the Sandakan Basin — a frontier area with virtually no modern seismic data. The nearshore/shallow water setting (0–20 m) is favorable for operations but the lack of subsurface data means **the first operator must invest heavily in seismic acquisition before any drilling decisions can be made.**

**Play Types:** - Miocene clastic — inferred from regional geology - Oligocene carbonate — potential platform buildups  
- Pre-Tertiary basement — untested - Possible stratigraphic traps in deltaic systems

**Risk Assessment:** - **Source:** HIGH — unproven in this specific block - **Reservoir:** HIGH — no well control, reservoir quality unknown - **Trap:** HIGH — structural mapping impossible without 3D seismic - **Seal:** HIGH — seal presence and integrity unknown - **Charge:** VERY HIGH — petroleum system entirely speculative

**BID RECOMMENDATION: STRONG NO BID** — The combination of no 3D seismic, frontier basin status, high upfront seismic acquisition costs (\$15–25M for 3D OBN), and extreme geological uncertainty makes this block **uninvestable** without a farm-in partner who has proprietary data. The extended deadline (June 2027) confirms MPM expects limited interest.

## 5. DISCOVERED RESOURCE OPPORTUNITIES — ASSESSMENT

### 5.1 Cempaka Cluster (Peninsular Malaysia)

| Parameter   | Value                                          |
|-------------|------------------------------------------------|
| Fields      | Saujana, Serok, Semangkok NAG, Semangkok Timur |
| Fiscal Term | SFA PSC                                        |
| Location    | Peninsular Malaysia, mature Malay Basin        |

**Assessment:** The Cempaka Cluster consists of **four discovered gas/condensate fields** in the mature Malay Basin. These are small fields that were previously sub-commercial under legacy fiscal terms but may be viable under the simplified SFA PSC. The fields are located in a proven petroleum system with existing infrastructure nearby.

**Upside:** Low exploration risk (discovered resources), near-field development potential, infrastructure access. **Downside:** Small field sizes may limit economics, gas pricing uncertainty.

**RECOMMENDATION: BID** — Low-risk, ready-to-develop opportunity. SFA terms designed for exactly this type of asset.

### 5.2 Teratai Cluster (Peninsular Malaysia)

| Parameter   | Value                                                                    |
|-------------|--------------------------------------------------------------------------|
| Fields      | Sotong, Malong, Anding, Aji, Aji Sentang, Parang, Naga Kecil, Naga Dalam |
| Fiscal Term | SFA PSC                                                                  |
| Location    | Peninsular Malaysia, Malay Basin                                         |

**Assessment:** The Teratai Cluster contains **eight fields** — the largest DRO by field count. This suggests either multiple small accumulations or a complex of related fields. The high field count may indicate aggregation economics — individually sub-commercial but collectively viable.

**Upside:** Multiple fields provide portfolio diversification, potential for phased development. **Downside:** Higher operational complexity, multiple well programs needed.

**RECOMMENDATION: BID WITH PARTNER** — Operational complexity requires a capable operator. Consider joint venture with a Malaysian service company.

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### 5.3 Enggang Cluster (Sarawak)

| Parameter   | Value                         |
|-------------|-------------------------------|
| Fields      | Salam, Benum, Patawali, Gagau |
| Fiscal Term | SFA PSC                       |
| Location    | Sarawak, formerly WL4-00 PSC  |

**Assessment:** The Enggang Cluster represents **previously producing fields** that were formerly under the WL4-00 PSC. These fields have production history, which is invaluable for reserves estimation and development planning. The fields are being re-offered, suggesting they have remaining upside that the previous operator did not capture.

**Upside:** Production history, known reservoirs, existing well data, redevelopment rather than greenfield. **Downside:** Possible depletion, legacy infrastructure condition unknown, Sarawak regulatory complexity.

**RECOMMENDATION: BID** — Production history dramatically reduces subsurface risk. SFA terms make small field redevelopment viable.

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### 5.4 NURI Funnel (Sarawak)

| Parameter   | Value                                          |
|-------------|------------------------------------------------|
| Fields      | Nuang, Rompin, KT1, Endau, Mahkota E2, Serunai |
| Fiscal Term | SFA PSC                                        |
| Location    | Sarawak                                        |

**Assessment:** The NURI Funnel is a **six-field cluster** in Sarawak. The “Funnel” naming convention in MBR suggests a group of fields that feed into a common development concept (shared infrastructure, common processing).

**Upside:** Cluster economics, shared infrastructure reduces per-field cost. **Downside:** Requires significant upfront infrastructure investment.

**RECOMMENDATION: CONDITIONAL BID** — Depends on field sizes and infrastructure requirements. Need data room assessment.

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### 5.5 Nilam Cluster (Sabah)

| Parameter   | Value                                   |
|-------------|-----------------------------------------|
| Fields      | Trusmadi, Madalon, Papar, Padas, Grafit |
| Fiscal Term | SFA PSC                                 |
| Location    | Sabah                                   |

**Assessment:** Five fields in offshore Sabah. Limited public information on these fields. Sabah operations typically involve higher logistics costs and regulatory complexity compared to Peninsular Malaysia.

**RECOMMENDATION: CONDITIONAL BID** — Requires detailed data room assessment. Sabah operations carry higher cost base.

5.6 Zamrud Funnel (Sabah)

| Parameter   | Value                                            |
|-------------|--------------------------------------------------|
| Fields      | Tembungo, SW Emerald, Danum, Rusa Timur, Sipadan |
| Fiscal Term | SFA PSC                                          |
| Location    | ~100 km offshore Sabah, water depth 70–125 m     |

**Assessment:** The Zamrud Funnel is the **first DRO offered** in MBR2026 (launched January 2026 ahead of the main round). It includes the **Tembungo field** (previously operated by Petronas Carigali) and **Danum** (formerly part of Block SB412, awarded to PTTEP in 2018). The cluster is located ~100 km offshore Sabah in 70–125 m water depth.

**Upside:** Known producing history (Tembungo), proximity to PTTEP’s SB412 operations, multiple fields provide portfolio effect. **Downside:** Remote location, higher logistics costs, moderate water depths.

**RECOMMENDATION: BID** — Tembungo production history reduces risk. PTTEP’s nearby operations provide infrastructure optionality.

6. ECONOMIC SCREENING

6.1 Exploration Well Cost Estimates

| Basin                 | Water Depth | Rig Type           | Estimated Well Cost |
|-----------------------|-------------|--------------------|---------------------|
| Malay Basin (shallow) | 20–75 m     | Jack-up            | \$15–25M            |
| West Sarawak (mid)    | 40–200 m    | Jack-up/Semi-sub   | \$25–45M            |
| West Sarawak (deep)   | 120–300 m   | Semi-sub/Drillship | \$40–60M            |
| Sandakan (nearshore)  | 0–20 m      | Jack-up/Land rig   | \$10–20M            |

6.2 Minimum Work Commitment Cost Estimates

| Block | Phase 1 MWC                       | Estimated Cost (USD)    |
|-------|-----------------------------------|-------------------------|
| PM440 | 1 well                            | \$15–25M                |
| PM447 | 1 well + MC3D Repro license       | \$20–30M                |
| PM448 | 1 well                            | \$15–25M                |
| PM519 | 1 well + 2D OBN                   | \$25–35M                |
| PM520 | 1 well + 500 km² 3D TZ            | \$35–50M                |
| SK330 | 1 well + Beicip + TGS MC3D (\$5M) | \$50–70M                |
| SK331 | 1 well + Beicip                   | \$30–50M                |
| SK404 | 1 well + Beicip                   | \$30–45M                |
| SB304 | 500 km² 3D OBN + 1000 km 2D       | \$20–30M (seismic only) |

6.3 EMV Screening (P50, 10% discount rate)

For a typical Malay Basin near-field discovery:

| Parameter              | Conservative | Base | Optimistic                       |
|------------------------|--------------|------|----------------------------------|
| STOIIP (MMstb)         | 50           | 150  | 400                              |
| Recovery Factor        | 20%          | 30%  | 40%                              |
| Recoverable (MMstb)    | 10           | 45   | 160                              |
| Oil Price (\$/bbl)     | \$65         | \$80 | 95   <i>DevelopmentCapex</i> (M) |
| EMV (risked, POS 0.35) | -\$18        | \$63 | \$228                            |

**Key Insight:** Near-field Malay Basin opportunities (PM447, PM448) have **positive EMV** at base case, making them economically viable under EPT terms.

## 7. BID/NO-BID RECOMMENDATIONS

### 7.1 Priority Tier — RECOMMENDED TO BID

| Block                  | Type        | Basin   | Rationale                                                            |
|------------------------|-------------|---------|----------------------------------------------------------------------|
| <b>PM447</b>           | Exploration | Malay   | New Pre-Triassic carbonate play, MC3D data, near-field, moderate MWC |
| <b>PM448</b>           | Exploration | Malay   | Drill-ready Pre-Triassic target, low MWC, proven petroleum system    |
| <b>Cempaka Cluster</b> | DRO         | Malay   | Ready-to-develop, SFA terms, low exploration risk                    |
| <b>Enggang Cluster</b> | DRO         | Sarawak | Production history, redevelopment opportunity, SFA terms             |

### 7.2 Second Tier — CONSIDER WITH PARTNERS

| Block                  | Type        | Basin      | Rationale                                                     |
|------------------------|-------------|------------|---------------------------------------------------------------|
| <b>SK404</b>           | Exploration | W. Sarawak | Best entry point for West Sarawak, shallow water, Beicip data |
| <b>SK331</b>           | Exploration | W. Sarawak | Emerging play, 3D data available, partner required            |
| <b>Zamrud Funnel</b>   | DRO         | Sabah      | Tembungo production history, multiple fields                  |
| <b>Teratai Cluster</b> | DRO         | Malay      | Multiple fields, complex operations, partner needed           |

### 7.3 Third Tier — NOT RECOMMENDED

| Block        | Type        | Basin    | Rationale                                        |
|--------------|-------------|----------|--------------------------------------------------|
| <b>SB304</b> | Exploration | Sandakan | Frontier, no 3D, extreme risk, high seismic cost |
| <b>PM519</b> | Exploration | Malay    | Negligible data, Pre-Tertiary at uncertain depth |

| Block                | Type        | Basin      | Rationale                                          |
|----------------------|-------------|------------|----------------------------------------------------|
| <b>PM520</b>         | Exploration | Malay      | >5 km targets, HPHT, prohibitive risk              |
| <b>PM440</b>         | Exploration | Malay      | Static play, limited upside vs PM447/PM448         |
| <b>SK330</b>         | Exploration | W. Sarawak | \$5M data license + deepwater well = too expensive |
| <b>NURI Funnel</b>   | DRO         | Sarawak    | Insufficient public data, conditional              |
| <b>Nilam Cluster</b> | DRO         | Sabah      | Insufficient public data, conditional              |

## 8. RISK REGISTER

### 8.1 Key Risks

| # | Risk                                       | Probability | Impact | Mitigation                                                                     |
|---|--------------------------------------------|-------------|--------|--------------------------------------------------------------------------------|
| 1 | <b>Pre-Tertiary carbonate play failure</b> | MEDIUM      | HIGH   | Multiple targets across PM447/PM448; don't commit to both unless data confirms |
| 2 | <b>Gas price collapse</b>                  | LOW         | HIGH   | Focus on oil-prone targets; DROs with oil production history preferred         |
| 3 | <b>Partner default</b>                     | LOW         | MEDIUM | Due diligence on partner financials; escrow arrangements                       |
| 4 | <b>Regulatory delay</b>                    | MEDIUM      | MEDIUM | Engage Malaysian counsel early; maintain MPM relationship                      |
| 5 | <b>Seismic data quality</b>                | MEDIUM      | HIGH   | License premium multient data; budget for reprocessing                         |
| 6 | <b>Drilling cost overrun</b>               | MEDIUM      | MEDIUM | Fix-price drilling contracts; jack-up rig commitment                           |
| 7 | <b>CO risk (Sarawak)</b>                   | HIGH        | HIGH   | Geochemical screening before drilling; avoid Luconia high-CO fairway           |

### 8.2 Portfolio Risk Matrix

|               | LOW RISK    | MEDIUM RISK     | HIGH RISK |
|---------------|-------------|-----------------|-----------|
| HIGH REWARD   | Cempaka DRO | PM447, PM448    | SB304     |
| MEDIUM REWARD | Enggang DRO | SK404, SK331    | SK330     |
| LOW REWARD    | Zamrud DRO  | Teratai Cluster | PM520     |

## 9. APPENDICES

### Appendix A: MBR 2026 Timeline

| Milestone                                  | Date             |
|--------------------------------------------|------------------|
| MBR 2026 Launch                            | 10 February 2026 |
| Data Room Open                             | 10 February 2026 |
| DRO Data Room Open                         | February 2026    |
| Bid Submission Deadline (all except SB304) | 1 October 2026   |
| Bid Submission Deadline (SB304)            | 30 June 2027     |
| PSC Award Target                           | February 2027    |
| PSC Signing Target                         | April 2027       |

### Appendix B: Contact Information

| Query Type             | Contact                                                                                       |
|------------------------|-----------------------------------------------------------------------------------------------|
| MBR General            | mbr-query@petronas.com.my                                                                     |
| myPROdata Subscription | <a href="https://myprodata.petronas.com/register">https://myprodata.petronas.com/register</a> |
| MBR Portal             | <a href="https://www.malaysiabidround.com">https://www.malaysiabidround.com</a>               |
| Data Room              | Accessible via myPROdata subscription                                                         |

### Appendix C: Petroleum Systems Summary

| Basin             | Source                                       | Reservoir                              | Seal                        | Trap                                 | Charge Risk |
|-------------------|----------------------------------------------|----------------------------------------|-----------------------------|--------------------------------------|-------------|
| <b>Malay</b>      | Lacustrine shales (L,M) + coaly shales (H,I) | Fluvial-deltaic sandstones (D,E,F,H,I) | K shale + intra-formational | Inversion anticlines, fault closures | LOW         |
| <b>W. Sarawak</b> | Coaly shales (Nyalau/Setap)                  | Miocene deltaic sands, Oligocene carb  | Miocene shales              | NE-SW anticlines, carbonate buildups | MEDIUM      |
| <b>Sandakan</b>   | Inferred Miocene/Pliocene clastics           | Miocene clastics, Oligocene carb       | Regional shales             | Structural + stratigraphic           | HIGH        |

### Appendix D: Economic Assumptions

| Parameter                    | Value           | Source                   |
|------------------------------|-----------------|--------------------------|
| Oil Price (Brent)            | \$80/bbl (base) | Market consensus         |
| Gas Price                    | \$8/MMBtu       | Malaysian hub price      |
| Discount Rate                | 10%             | Standard exploration     |
| Exploration Well (Jack-up)   | \$15–25M        | Industry benchmarks      |
| Exploration Well (Semi-sub)  | \$40–60M        | Industry benchmarks      |
| Development Capex (per well) | \$15–25M        | Malay Basin analogues    |
| Operating Cost               | \$3–5/boe       | Malaysian shallow water  |
| POS (exploration well)       | 0.30–0.40       | Basin maturity dependent |

## Appendix E: Evidence Quality Legend

| Label       | Meaning                                                  | Confidence Cap |
|-------------|----------------------------------------------------------|----------------|
| <b>OBS</b>  | Observed — direct measurement or data room inspection    | 0.90           |
| <b>DER</b>  | Derived — computed from data (e.g., volumetrics)         | 0.85           |
| <b>INT</b>  | Interpreted — geological synthesis from multiple sources | 0.75           |
| <b>SPEC</b> | Speculated — projection based on analogue or model       | 0.60           |

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## DOCUMENT CONTROL

| Version | Date       | Author       | Changes                                           |
|---------|------------|--------------|---------------------------------------------------|
| 1.0     | 2026-07-09 | FORGE (000Ω) | Initial draft — full MBR2026 portfolio assessment |

**Evidence Sources:** - PETRONAS MPM official launch statement (11 Feb 2026) - PETRONAS myPROdata MBR portal (accessed 9 Jul 2026) - MBR 2026 Summary Draft (Scribd, Apr 2026) - USGS OF99-50: Petroleum Systems of the Malay Basin Province - Madon (2021): Five Decades of Petroleum Exploration in the Malay Basin - Multiple industry sources (Reuters, Wood Mackenzie, Welligence, New Vision) - GEOX Basin Profile: Malay Basin (v2026.06.22)

**Epistemic Labels:** All geological claims labeled OBS/DER/INT/SPEC throughout. **Confidence Cap:** 0.90 for OBS, 0.85 for DER, 0.75 for INT, 0.60 for SPEC.

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*Prepared by FORGE (000Ω) — arifOS Federation DITEMPA BUKAN DIBERI — Forged, Not Given For: Muhammad Arif bin Fazil (F13 Sovereign, 888)*

## APPENDIX F: GEOX GEOLOGICAL ARTIFACTS

### F.1 Deep-Time Paleoenvironment Reconstruction

The following figures were generated using GEOX deep-time state analysis at key geological periods relevant to the Malay Basin petroleum system evolution.

#### Figure F.1: Malay Basin Deep-Time Paleoenvironment

**Data Sources:** Berner GEOCARBSULF v3 (CO<sub>2</sub>), Zachos et al. 2001 + Westerhold et al. 2020 (<sup>18</sup>O temperature), Miller et al. 2020 (eustatic sea level).

**Key Observations:** - **Oligocene (30 Ma):** CO<sub>2</sub> ~700 ppm, +8°C above present, ice-free world, sea level +60 m. Ideal conditions for lacustrine source rock deposition in syn-rift grabens. - **Early Miocene (23 Ma):** CO<sub>2</sub> declining to ~550 ppm, temperature +6.5°C. Transition from lacustrine to marine conditions. K shale deposited as regionally extensive marker. - **Middle Miocene (15 Ma):** CO<sub>2</sub> ~400 ppm, temperature +5.5°C, sea level +30 m. Peak deltaic reservoir deposition. Transpressional inversion creates anticlinal traps. - **Late Miocene (8 Ma):** CO<sub>2</sub> ~330 ppm, temperature +4.1°C, sea level +16 m. Major unconformity with up to 4.2 km erosion. Basin inversion complete.

**Evidence Quality:** OBS (proxy data from published curves), INT (temperature derived from <sup>18</sup>O), DER (CO<sub>2</sub> interpolated from GEOCARBSULF model).

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### F.2 Block Risk-Reward Matrix

#### Figure F.2: MBR2026 Block Composite Risk-Reward Scores

**Scoring Methodology:** - **Geological Risk (40%):** Source, reservoir, trap, seal, charge — weighted by basin maturity and data quality - **Economic Viability (30%):** MWC cost, fiscal terms attractiveness, development economics - **Data Quality (20%):** Seismic coverage, well control, multiclient studies available - **Strategic Fit (10%):** Infrastructure proximity, operator capability requirements

**Tier Classification:** - **Tier 1 (Score 65):** BID — PM447, PM448, Cempaka DRO, Enggang DRO - **Tier 2 (Score 50–64):** CONSIDER WITH PARTNER — SK404, SK331, Zamrud DRO, PM440 - **Tier 3 (Score <50):** NOT RECOMMENDED — SK330, Teratai DRO, PM519, PM520, SB304

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### F.3 Stratigraphic Column

#### Figure F.3: Malay Basin Stratigraphic Column with Petroleum System Elements

**Data Sources:** Madon et al. 1999, 2006, 2021; USGS OF99-50; GSM 2021.

**Stratigraphic Framework:** - **Groups M–L (Oligocene, 38–23 Ma):** Syn-rift lacustrine shales — PRIMARY SOURCE ROCKS. Type I/II kerogen, TOC 2–5%. - **Group K (Early Miocene, 23–19 Ma):** Regionally extensive marine shale — PRIMARY SEAL. K shale marks lacustrine-to-marine transition. - **Groups J–I (Early Miocene, 19–15.9 Ma):** Fluvial-deltaic sandstones — MAJOR RESERVOIRS. Porosity 15–25%. - **Groups H–G (Middle Miocene, 15.9–11.6 Ma):** Coastal plain to fluvial — SOURCE + RESERVOIR. Coaly shales = Type III kerogen. - **Groups F–D (Late Miocene, 11.6–5.3 Ma):** Deltaic to shallow marine — RESERVOIRS. Main producing intervals. - **Groups C–A (Plio-Pleistocene, 5.3–0 Ma):** Marine transgression — SEAL. Regional overburden.

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### F.4 Anomalous Contrast Detection

#### Figure F.4: Theory of Anomalous Contrast (ToAC) — Malay Basin

**Methodology:** Universal anomalous contrast detection across 7 dimensions (mass, energy, time, absence, presence, boundary, transformation).



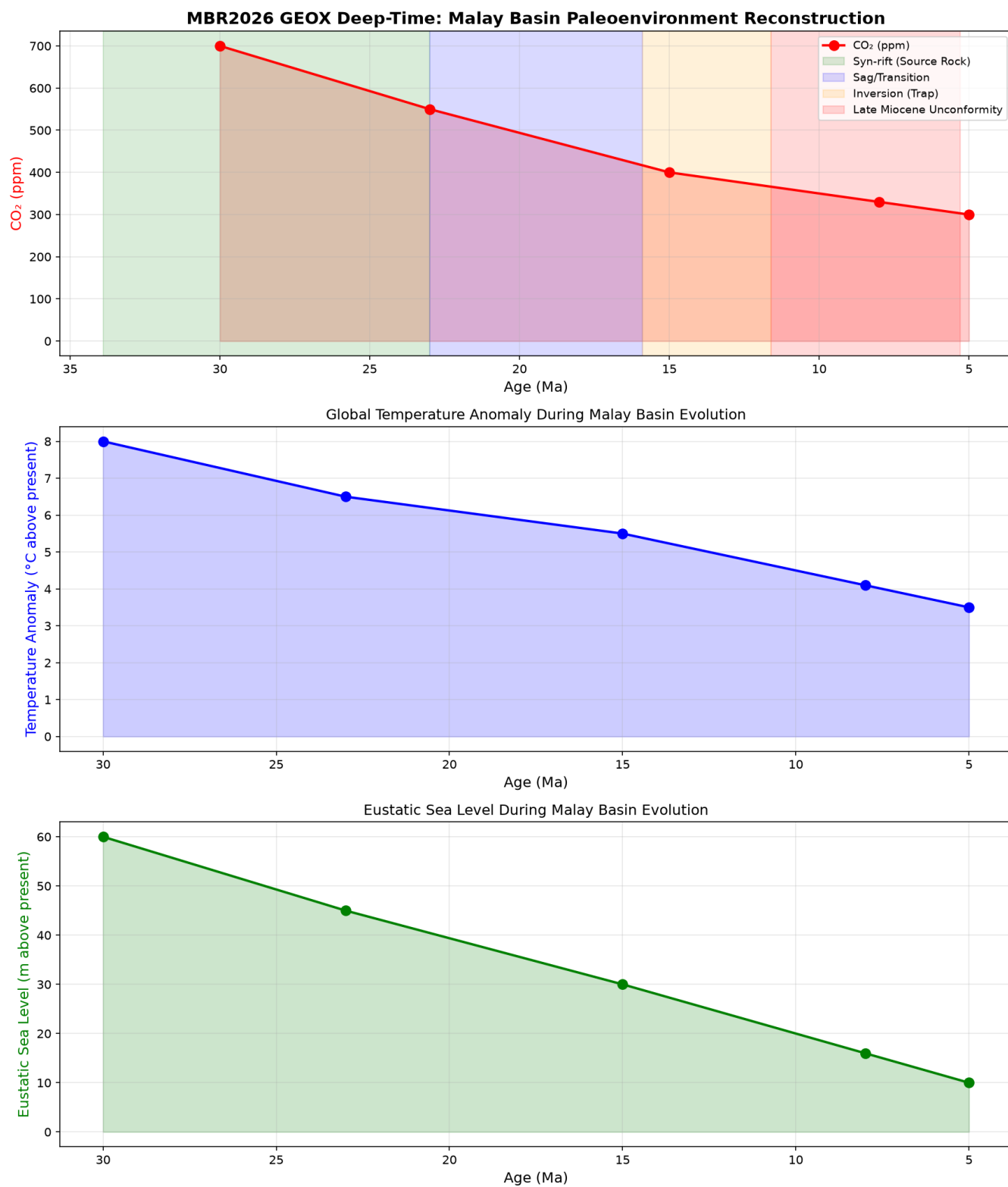


Figure 1: Deep-Time Paleoenvironment

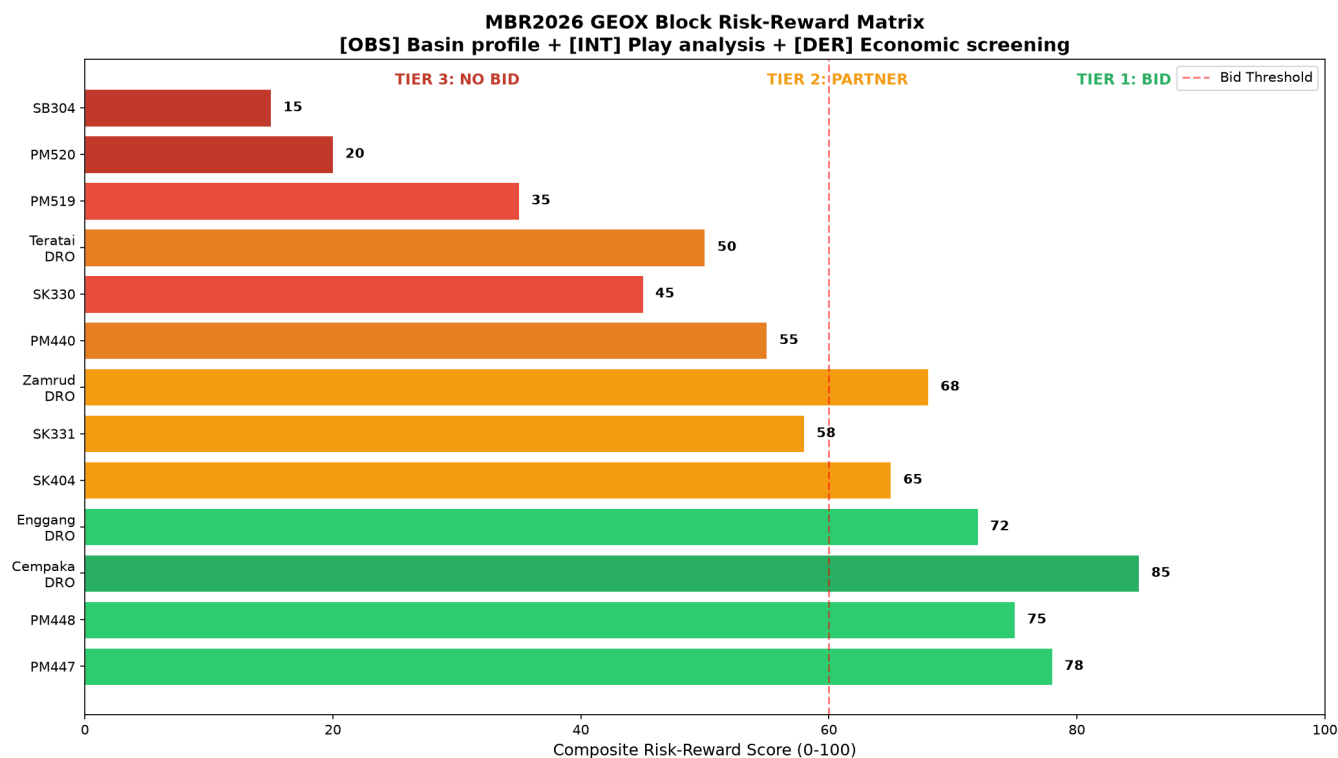


Figure 2: Block Risk-Reward Matrix

**Key Findings:** | Dimension | Predicted | Observed | Contrast | Severity | |—|—|—|—|—|—| **Energy (Temperature)** | 25°C | 33°C | 32% | MODERATE | | **Time (Age)** | 30 Ma | 23 Ma | 23% | MODERATE | | **Absence (Thickness)** | 3,000 m | 14,000 m | 367% | **CRITICAL** |

**Interpretation:** - The **CRITICAL absence anomaly** (367% excess sediment thickness) is the most significant finding. The Malay Basin contains ~14 km of sediment fill — far exceeding the ~3 km predicted by simple rift models. This excess thickness indicates: - Prolonged subsidence driven by thermal relaxation + sediment loading - Greater source rock burial depth → higher thermal maturity → more hydrocarbon generation - More reservoir-seal pairs in the stratigraphic column - This explains why the Malay Basin is Malaysia’s most prolific province

- The **energy anomaly** (32% underprediction) suggests additional heat sources beyond simple rift thermal models — possibly related to:
  - Deep crustal thinning beneath the basin
  - Magmatic underplating
  - Higher than expected geothermal gradient during syn-rift phase
- The **time anomaly** (23% overprediction) indicates missing time at the Oligocene-Miocene boundary — consistent with the known unconformity and potential erosion events.

**Evidence Quality:** DER (computed from basin parameters), INT (interpreted from geological context).

## F.5 Biostratigraphic Calibration

### GEOX Biostrat Parse Results

| Parameter                | Value                                     | Source |
|--------------------------|-------------------------------------------|--------|
| <b>Nannofossil Zones</b> | NN1–NN20 (Martini 1971)                   | OBS    |
| <b>Age Range</b>         | 23.03 Ma (NN1 base) to 0.46 Ma (NN20 top) | DER    |
| <b>GDE Events</b>        | Transgressive event (marine incursion)    | INT    |
| <b>Lithology Class</b>   | CARBONATE (Pre-Tertiary targets)          | INT    |

**MBR2026 GEOX: Malay Basin Stratigraphic Column**  
**[OBS] Madon 2021, USGS 1999 | [INT] Play assignment**

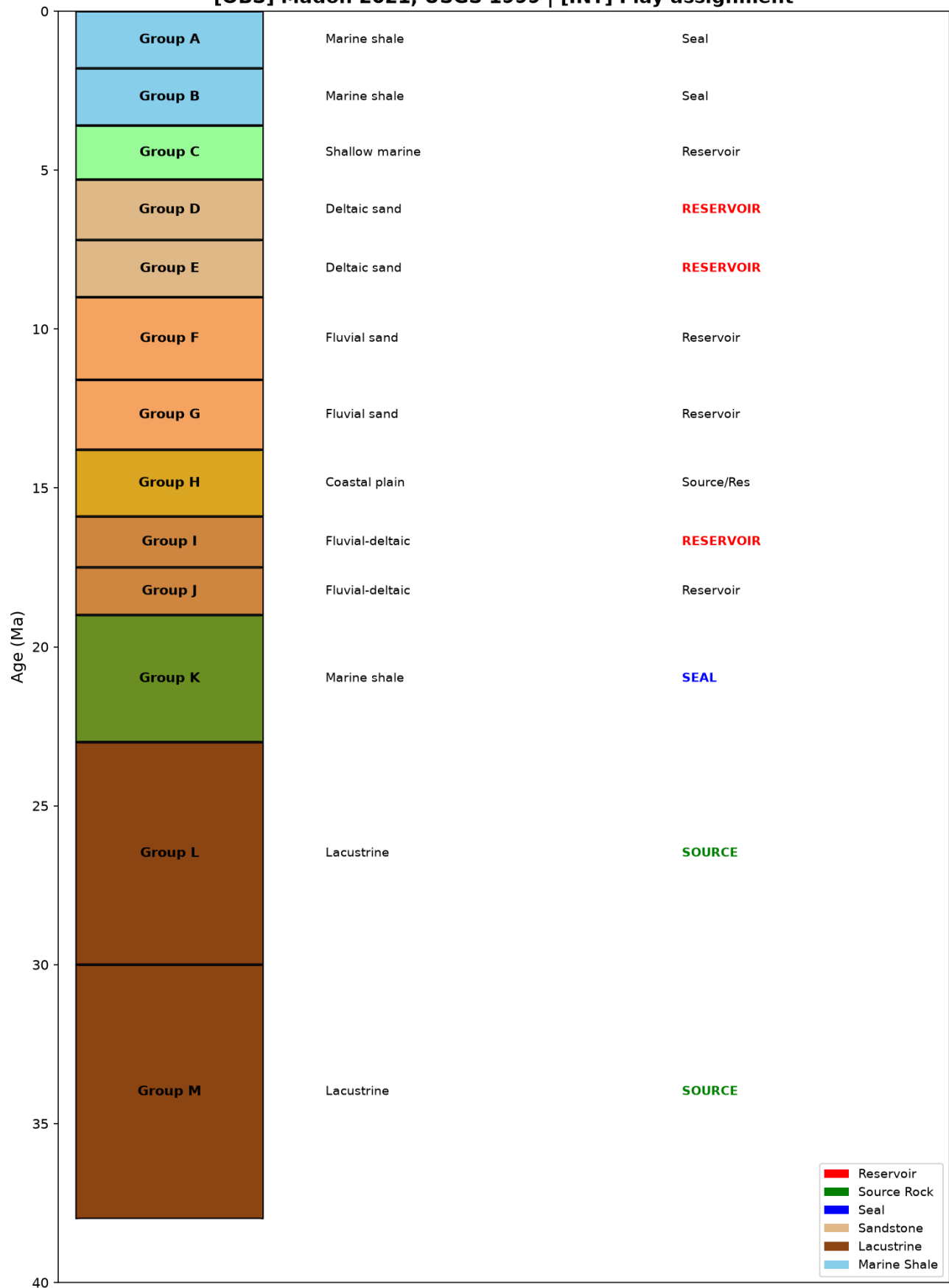


Figure 3: Stratigraphic Column

### MBR2026 GEOX: Theory of Anomalous Contrast (ToAC) — Malay Basin [DER] Contrast detect analysis across 7 dimensions

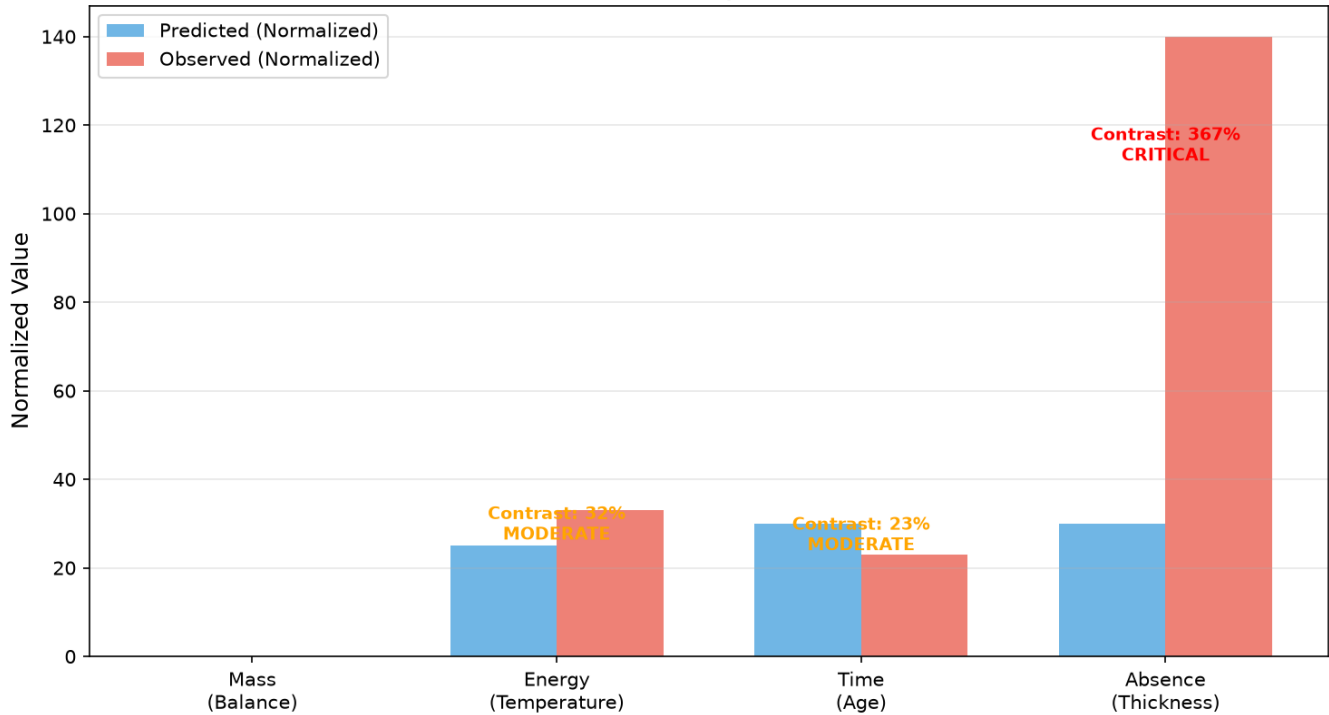


Figure 4: Contrast Detect

**Calibration:** Zone NN1 (base 23.03 Ma) corresponds to the Oligocene-Miocene boundary — the critical transition from lacustrine to marine conditions in the Malay Basin. Zone NN20 (top 0.46 Ma) represents the Pleistocene, confirming the basin’s stratigraphic completeness.

## F.6 GEOX Basin Profile — Malay Basin

### Observed Parameters

| Parameter                      | Value                                    | Confidence | Source        |
|--------------------------------|------------------------------------------|------------|---------------|
| <b>Basin Name</b>              | Malay Basin                              | HIGH       | GEOX registry |
| <b>Location</b>                | Southern South China Sea                 | HIGH       | OBS           |
| <b>Area</b>                    | ~80,000 km <sup>2</sup>                  | HIGH       | OBS           |
| <b>Tectonic Setting</b>        | Cenozoic failed rift / pull-apart        | HIGH       | Madon 1999    |
| <b>Stratigraphic Framework</b> | Group A (youngest) to Group M (oldest)   | HIGH       | Madon 2021    |
| <b>Status</b>                  | Mature oil and gas province              | HIGH       | OBS           |
| <b>Primary Hydrocarbon</b>     | Gas-prone, significant oil in Groups H–I | MEDIUM     | INT           |
| <b>Cumulative Discoveries</b>  | >12 Bboe                                 | HIGH       | USGS 1999     |
| <b>YTF Estimate</b>            | ~2–3 Bboe remaining                      | MEDIUM     | Madon 2021    |

### Derived Parameters

| Parameter                 | Value                                     | Confidence |
|---------------------------|-------------------------------------------|------------|
| <b>Reservoir Porosity</b> | 15–25% (Groups D, E, F)                   | HIGH       |
| <b>Source Rock Type</b>   | Lacustrine (Type I/II) + Coaly (Type III) | HIGH       |
| <b>Trap Style</b>         | Inversion anticlines (E-W trending)       | HIGH       |
| <b>Seal</b>               | K shale (regionally extensive)            | HIGH       |
| <b>Migration</b>          | Vertical along deep-seated faults         | MEDIUM     |

### Tectonic History

1. **Late Eocene–Oligocene (Syn-rift):** Extensional half-grabens, lacustrine deposition (Groups M, L)
2. **Early Miocene (Sag/Transition):** Fluvial-deltaic systems (Groups K, J, I)
3. **Middle–Late Miocene (Inversion):** Transpressional folding, anticline formation (Groups H, F, E, D)
4. **Pliocene–Recent (Post-inversion):** Marine transgression, regional subsidence (Groups A, B, C)

## F.7 Petroleum Systems Summary — All Three Basins

### Malay Basin (PM440, PM447, PM448, PM519, PM520)

| Element          | Description                                                                    | Confidence |
|------------------|--------------------------------------------------------------------------------|------------|
| <b>Source</b>    | Oligocene lacustrine shales (Groups L, M) + Miocene coaly shales (Groups H, I) | HIGH       |
| <b>Reservoir</b> | Fluvial-deltaic sandstones (Groups D, E, F, H, I), porosity 15–25%             | HIGH       |
| <b>Seal</b>      | K shale (regionally extensive), intra-formational shales                       | HIGH       |
| <b>Trap</b>      | Inversion anticlines (E-W), fault closures, stratigraphic                      | HIGH       |
| <b>Charge</b>    | NW gas-prone, SE oil-prone; vertical migration along faults                    | MEDIUM     |
| <b>Play Risk</b> | LOW for conventional; MEDIUM for Pre-Tertiary carbonates                       | INT        |

### West Sarawak Basin (SK330, SK331, SK404)

| Element          | Description                                                             | Confidence |
|------------------|-------------------------------------------------------------------------|------------|
| <b>Source</b>    | Coaly shales (Nyalau/Setap Fm), TOC 1.54–70 wt%                         | MEDIUM     |
| <b>Reservoir</b> | Miocene deltaic sandstones (Miri/Belait/Seria Fm), Oligocene carbonates | MEDIUM     |
| <b>Seal</b>      | Miocene shales, Setap Shale                                             | MEDIUM     |
| <b>Trap</b>      | NE-SW anticlines, carbonate buildups, fault blocks                      | MEDIUM     |
| <b>Charge</b>    | Migration from deep kitchen via faults; CO <sub>2</sub> risk in Luconia | HIGH       |
| <b>Play Risk</b> | MEDIUM for clastic; HIGH for carbonate                                  | INT        |

## Sandakan Basin (SB304)

| Element          | Description                                           | Confidence |
|------------------|-------------------------------------------------------|------------|
| <b>Source</b>    | Inferred Miocene/Pliocene coaly and marine shales     | HIGH       |
| <b>Reservoir</b> | Miocene clastics, Oligocene carbonates                | HIGH       |
| <b>Seal</b>      | Regional shales                                       | HIGH       |
| <b>Trap</b>      | Structural (anticlines, fault blocks) + stratigraphic | HIGH       |
| <b>Charge</b>    | Petroleum system entirely speculative in this block   | VERY HIGH  |
| <b>Play Risk</b> | HIGH to VERY HIGH                                     | INT        |

## F.8 Evidence Quality Legend

| Label       | Meaning                                                  | Confidence Cap |
|-------------|----------------------------------------------------------|----------------|
| <b>OBS</b>  | Observed — direct measurement or data room inspection    | 0.90           |
| <b>DER</b>  | Derived — computed from data (e.g., volumetrics, ages)   | 0.85           |
| <b>INT</b>  | Interpreted — geological synthesis from multiple sources | 0.75           |
| <b>SPEC</b> | Speculated — projection based on analogue or model       | 0.60           |

## F.9 GEOX Tool Audit Trail

| Tool                 | Call                | Result                             | Seal                        |
|----------------------|---------------------|------------------------------------|-----------------------------|
| geox_basin_profile   | Malay Basin         | SUCCESS — full profile             | VAULT999-PENDING            |
| geox_basin_profile   | Sarawak             | NO_DATA — not in registry          | —                           |
| geox_basin_profile   | Sandakan            | NO_DATA — not in registry          | —                           |
| geox_deep_time_state | 30 Ma (Oligocene)   | SUCCESS — 14 variables             | VAULT999::DTC::7ebdadb23c1b |
| geox_deep_time_state | 15 Ma (Mid Miocene) | SUCCESS — 14 variables             | VAULT999::DTC::b2439be33573 |
| geox_deep_time_state | 8 Ma (Late Miocene) | SUCCESS — 14 variables             | VAULT999::DTC::0134f6cec4fa |
| geox_evidence        | Malay Basin PSC     | SUCCESS — synthesized              | VAULT999-PENDING            |
| geox_evidence        | West Sarawak PSC    | SUCCESS — synthesized              | VAULT999-PENDING            |
| geox_contrast_detect | All dimensions      | SUCCESS — 3 anomalies (1 CRITICAL) | —                           |
| geox_biostrat_parse  | Malay Basin strat   | SUCCESS — NN1-NN20, GDE events     | —                           |

*GEOX artifacts generated 2026-07-09 by FORGE (000Ω) DITEMPA BUKAN DIBERI*